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Palaeomagnetic Studies in the British Caledonides—IV Lower Devonian Lavas of the Strathmore Region, Scotland

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Summary

Mean palaeomagnetic directions after dip-correction at 61 sites in lavas, ignimbrites and sediments in the Lower Devonian sequence of the Strathmore region, show a principal clustering around an axis $D = 42°/222°$ E, $I = -49°/+49°$ which are regarded as normal and reversed (N and R) polarity groups respectively. The frequency distribution of site mean directions is not simply bipolar, however, having subsidiary maxima around $D = 76°$, $I = +29°$ ('Group A') and $D = 162°$, $I = -35°$ ('Group B'). It is shown that the population can be subdivided into N and R Fisherian distributions (34 sites altogether) plus Groups A and B (18 sites together) and a random background of nine 'unclassified' sites. This analysis leads to a best estimate of the mean N and R directions corresponding to a palaeomagnetic pole 5° N 140° E ($d\psi = 5°$, $d\chi = 9°$). The primary (Lower Devonian) origin for the stable remanent direction, at least in the N, R, A and B groups is demonstrated by fold tests, a conglomerate test, igneous contact tests and by the recognition of a magnetic polarity stratigraphy consisting of at least eight N and R zones and at least five zones of 'A' or 'B' polarity. These latter groups are believed to record preferred intermediate directions of the geomagnetic field during polarity transitions. Remanent intensity is similar in the lavas in all polarity groups; it is believed to be the product of oxidation during cooling of the lavas. The more usual situation of lower intensity in rocks formed during polarity transitions is perhaps observed in intermediate polarity samples of the Highland Border Ignimbrite and may indicate an original TRM in these rocks.

A composite calculation of the mean N and R direction after AF cleaning based on this and previous studies in the eastern part of the Midland Valley of Scotland (50 sites altogether) gives $D = 41°$, $I = -40°$ ($\alpha_{95} = 6°$); the corresponding palaeomagnetic pole is 4° N, 140° E ($d\psi = 4°$, $d\chi = 7°$).

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1. Introduction

The Lower Old Red Sandstone rocks of the Midland Valley of Scotland consist mainly of sediments but include considerable thicknesses of lavas. The extrusives mostly range in composition between andesite and basalt but include subordinate rhyodacites and trachyandesites. The palaeomagnetism of these rocks was first studied in the Strathmore region by Stubbs (1958) who found directions of magnetization different from those determined from a study of the Old Red Sandstone sediments of the Anglo-Welsh cuvette (Creer 1957). This conflict is one of several concerning Devonian palaeomagnetic results throughout Europe, reviewed in Paper VI. Notably all well established palaeomagnetic poles from British Devonian rocks lie well to the west of contemporary poles derived from Norway and Russia. The British data comprise results from the Midland Valley *lavas* (Stubbs 1958; Embleton 1968; McMurry 1968, 1970), the flat lying Old Red Sandstone *sediments* of the Anglo-Welsh cuvette (Chamalaun & Creer 1964) and *intrusive* rocks of 'Lower Old Red Sandstone age' in the south-west highlands of Scotland (Briden 1970). Chamalaun & Creer (1964) showed that, apart from a minor high critical blocking temperature component revealed during thermal demagnetization, the NRM in the Anglo-Welsh Old Red Sandstone sediments is largely secondary, and unrelated to the Devonian magnetic field.

This study was undertaken to establish details of the British palaeomagnetic record using the detailed stratigraphic information derived from recent mapping by the Institute of Geological Sciences (Armstrong & Paterson 1970).

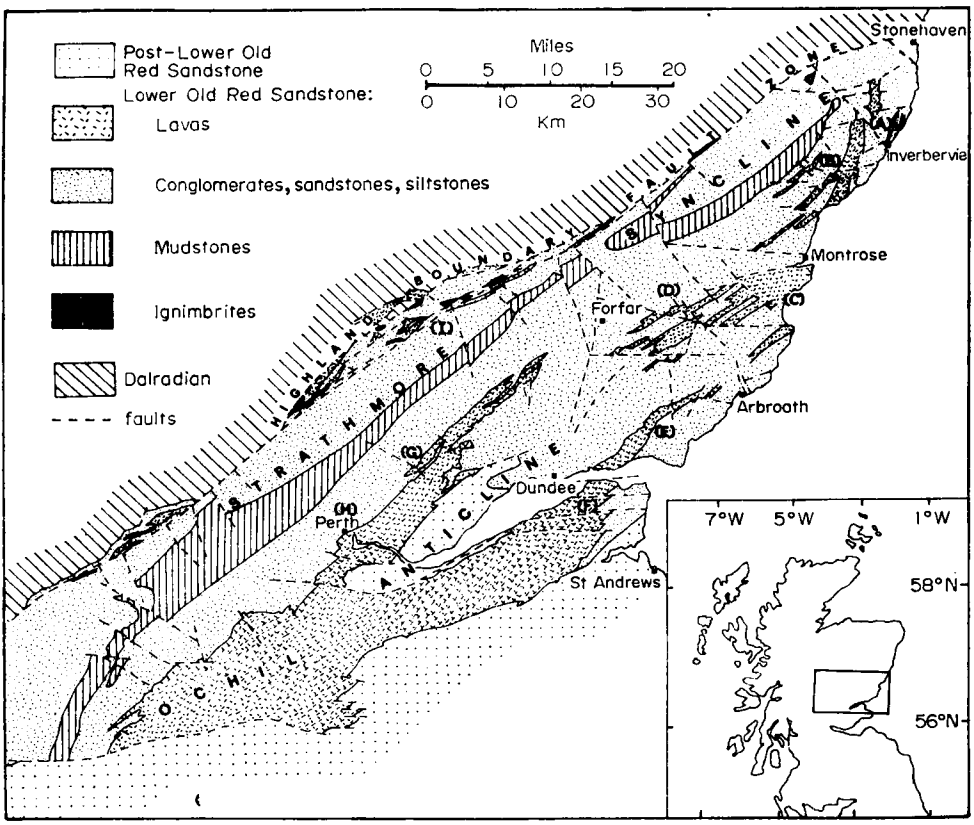


FIG. 1. Geological map of the Strathmore region of the Midland Valley of Scotland simplified after Armstrong & Paterson (1970). The letters in brackets refer to stratigraphic columns illustrated in Fig. 8 and discussed in Section 4.4.

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2. Geology

The Lower Old Red Sandstone rocks of the Strathmore region are composed of medium to coarse grained sandstones with thick horizons of conglomerates and lavas; shales and mudstones are locally important. The outcrop covers an area of some 4000 sq. km in the north-eastern part of the Midland Valley (Fig. 1) and is bordered in the north by the Highland Boundary Fault zone, and in the south by the Ochil Fault or overlain unconformably by Upper Old Red Sandstone. The rocks were folded in Middle Devonian times along a WSW–ENE strike to produce the Strathmore syncline and the Sidlaw anticline (Fig. 1), and considerable movement took place along the sub-parallel Highland Boundary Fault zone. The rocks have recently been described by Armstrong & Paterson (1970) within a stratigraphical scheme that retains the five groups distinguished by Campbell (1913) in Kincardineshire where the most complete sequence is present. The oldest strata were considered by Campbell (1911) on the basis of the fish and arthropod fauna to be Downtonian in age and overlie rocks of possibly Cambrian age near the Highland Boundary Fault. The major subdivisions are (Armstrong & Paterson 1970, Table 1):—

Group	Thickness	Stage
Strathmore	2000 m	Breconian
Garvock	1525 m	
Arbuthnott	2100 m	Dittonian
Crawton	670 m	
Dunnottar	1660 m	
Stonehaven	1550 m	Downtonian

All of the lower five groups contain volcanic rocks which have been sampled in the course of the present work. We follow Armstrong & Paterson (1970) in including the Downtonian within the Lower Old Red Sandstone, and hence within the Devonian.

The Stonehaven Group, which is confined to Kincardineshire, extends westwards from the coast at Stonehaven in a narrow outcrop flanking the Highland Boundary Fault on the almost vertical NW limb of the Strathmore syncline. The group comprises a lower (Cowie) and upper (Carron) formation both of which consist mainly of sandstone. A thin and impersistent andesite lava, first reported by Hutchinson (1928), occurs near the base of the Cowie Formation. Samples were taken from this lava (site 31) and also from the immediately underlying red sandstones (site 32).

The Dunnottar Group is composed largely of coarse conglomerate but includes at the top several flows of olivine basalt which have been assigned to the Tremuda Bay Volcanic Formation. This is well exposed at Tremuda Bay where two separate flows were sampled during the present study (sites 33 and 34). A third flow was sampled by McMurry (1968, site T1). At Tod Head Point a single lava flow, correlated with the Tremuda Bay lavas by Campbell (1913), was also sampled by McMurry (site D1).

The Crawton Group consists predominantly of coarse conglomerate much of which consists of material derived from the Highland region. Near the top it includes the Crawton Volcanic Formation which comprises a distinctive suite of olivine basalts and basic andesites containing large phenocrysts of feldspar. McMurry sampled the lavas at three localities on the coast (sites W1, B1 and H1) and further samples were taken during the present study at Bervie Brow (site 36). In the type section at Crawton Bay three flows of macroporphyritic lava are present. Samples were taken from the bottom flow (site 26), from each of the two units of the middle flow (sites 27 and 28) and one site (29) from the top flow. Seven samples were taken

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from boulders of the distinctive macroporphyritic lava in the immediately overlying conglomerate for a conglomerate test (site 30). Stubbs (1958) also sampled this formation and his sites have been repeated by McMurry (1968, sites C1, C2, C3 and C4). Embleton (1968) sampled the same rocks but discarded the results alleging that the magnetism was unstable. Rocks near the Highland Border which were previously described as dacite lavas (Campbell 1913) have recently been shown by Paterson & Harris (1969) to be ignimbrites and have been placed on petrographic grounds within a single formation. They are considered by Armstrong & Paterson (1970) to be at approximately the same stratigraphic horizon as the Crawton Volcanic Formation. Samples have been taken from seven sites at three localities (sites 105, 107, 133, 173, 174, 176 and 177).

The Arbuthnott Group has been subdivided into four diachronous formations by Armstrong & Paterson (1970). The oldest of these, the Johnshaven formation, consists of conglomerate and subordinate sandstone with several lava flows. Two of the four flows exposed in the section at Johnshaven harbour were sampled (24, 37) and also a thin basalt flow at Bervie Brow (35) near the base of the formation.

The overlying Montrose Volcanic Formation consists of lenticular lava bodies intercalated in the sediments of the Johnshaven and Dundee formations. On the north-west side of the Sidlaw anticline the formation is divided into the Ferryden, Dunnichen, St Cyrus and Morphie-Bruxie Hill members. A strike fault has brought the St Cyrus and Morphie members together south of Arbuthnott where they were sampled by Embleton (1968, sites L13, L14). On the south limb of the anticline lavas were sampled at Kirkton of Tealing (17), East Hills and West Hills (50, 61), and at Ethie Haven by McMurry (1968, site E1). These lava outcrops are believed to be equivalent to the Ferryden, Dunnichen, and St Cyrus members (Armstrong & Paterson 1970). Two other lavas within the St Cyrus member have been sampled at St Cyrus harbour (22, 23). The Dunnichen member was sampled at Dunnichen Hill (58, 59). Another lava, sampled at Knock Hill (sites 41, 42 and McMurry's site K1) was correlated by Campbell (1913) with rocks now assigned to the Morphie-Bruxie member, but has been placed in the middle of the Arbuthnott group by Armstrong & Paterson (1970) at about the same level as the St Cyrus member. Lavas of the Morphie-Bruxie Hill member were sampled at East Mathers (38, 39), near Glenbervie (43, McMurry's site M2) and at Bruxie Hill by Embleton (1968, site L10) and McMurry (site CB1). Lavas of the Ferryden member were sampled along the coast from Montrose to Lunan Bay by McMurry (sites U1, U2, F1 and N1) and during the present work at sites 63, 64 and also at sites 20 and 21 on Compass Hill.

The Ochil Volcanic Formation has an extensive outcrop on the NW limb of the Sidlaw anticline and in Perthshire and Angus forms the high ground of the Sidlaw Hills. These lavas were systematically sampled through their thickness at sites along a line from Pitrodie northwards to Collace (45, 46, 47, 48, 49, 14, 13, 53, 56, 57, 51, 52, 50, 9, 8). The formation was also sampled at other sites to the west of this traverse (16, 15, 10, 44) and at three sites near Perth (11, 12A and 12B) and two near Kinross (3, 4) on the SE limb of the Sidlaw anticline. On this flank of the major anticlinal structure the Ochil Volcanic Formation extends from near Kinross to the Dundee district where, at Broughty Ferry, the main body of lavas is underlain by a mixed sequence of lavas and intercalated sediments belonging to the Dundee Formation. Lavas of the Ochil Volcanic Formation have also been sampled in northern Fife (sites 18, 19), and Embleton (1968) sampled the formation in southern Fife (sites L3 and L4). The lavas of the Arbuthnott Group are intercalated on the steep north-west limb of the Strathmore syncline with a thick succession of sandstones and conglomerates but, because of extreme variability of the sequence, correlation with the SE limb of the syncline is uncertain. The Garvock Group is predominantly sandstones and conglomerates with a few thin lava flows which have been sampled during the present work at Laurencekirk (62) and Bervie Water (40). Volcanic rocks are

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absent from the uppermost (Strathmore) group of the Lower Old Red Sandstone in this area.

Altogether 67 new sites have been sampled from the Lower Old Red Sandstone rocks of north-east Scotland. Seven of these are from the Highland Border Ignimbrite, three are from sediments adjacent to lavas and the remainder are from lavas. All the lava sites of this collection were drilled on location and oriented by sunsights and/or magnetic bearings; the ignimbrite sites proved to be either too hard or too fractured for effective field drilling and they were hand sampled and cored in the laboratory.

Evans *et al.* (1971) have reported a mean age of 394 ± 5 My for three lavas from the Tayside area which are the approximate lateral equivalent of our sites 18 and 19, and assuming that the Lower Devonian interval lasted for 20 My it is suggested that the total time span covered by the palaeomagnetic collection from the Strathmore region may be of the order of 5 to 10 My.

3. Palaeomagnetic results

3.1 *NRM directions*

The total NRM directions are generally consistent within each site and only seven site mean directions show a grouping which is not significant at the 95 per cent confidence level (Fisher 1953; Watson 1956; Irving 1964). Site-mean data are summarized in Table 1 after correction for tilt. Normal and reversed directions of magnetization are discernible but they show a large scatter (Fig. 2(a)). After magnetic cleaning the directions of magnetization group into normal and reversed directions (*N*, *R*), intermediate directions (*A* and *B*) and unclassified directions, and these subdivisions are used throughout this discussion.

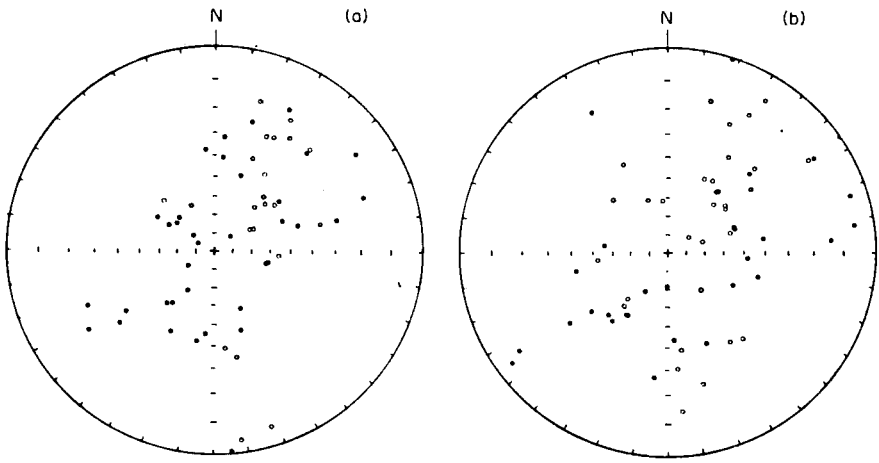


FIG. 2 (a) Site mean total NRM directions, and (b) site mean AF or thermally cleaned directions after correction for dip. Throughout this paper open symbols refer to upper hemisphere plots and closed symbols to the lower hemisphere.

3.2. *AF demagnetization*

Progressive AF demagnetization has been carried out on one or two pilot samples from each site. They typically show a steady fall in their moment with little change in remanent direction of NRM (Fig. 3(a)). The pilot samples of sites 3 and 10 (Fig. 3(b)) show a sharp decrease in moment on demagnetization. In the former case there is no change in the direction of the moment and it is suggested that all the NRM is aligned in a single direction, but in the latter case the total NRM is dominated by a low

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coercivity component close to the present geomagnetic field direction. The stable north-westerly direction is retained in the fraction with coercivity greater than 200 Oe. Pilot samples from sites 28, 29 and 31 retain more than 50 per cent of their NRM intensity after demagnetizing in 1000 Oe (Fig. 3(c)) and site 31 has 95 per cent of its total NRM remaining after demagnetization in 1600 Oe. These high coercivities may be due to the presence of haematite and this correlates with the observed highly oxidised state of these rocks (Section 4.2).

An optimum field for cleaning the remainder of the sites was based on the behaviour of the pilots as defined by Briden's 1972 Stability Index. These fields are mostly in the range 300–500 Oe, and after cleaning the number of non-significant site mean directions is reduced from eleven to six (Table 1).

The within-site scatter is decreased at most sites after AF demagnetization but deteriorates significantly for sites 3 and 46. This probably results from the bulk of the magnetization residing in low coercivity material. The total NRM mean directions of these two sites are used in the final computation of the overall mean although their mean directions after AF cleaning are not appreciably different. Sites 20, 26, 61 and 173 behaved in an unstable fashion during AF treatment and are no longer statistically acceptable. Sites 20 and 26 appear to have acquired an unstable component at a later geological time and site 61 has a remanence after cleaning close to the limits of detectability of the instrument. Site 64 was random throughout treatment.

3.3. *Thermal demagnetization results*

One sample from each site was thermally demagnetized in seven steps up to 630° C. In most cases there was little or no change in direction of remanence (Fig. 4(a)), and where there was some change the specimens reached a stable direction at temperatures of 300–500° C which in every case was close to that isolated by AF demagnetization. The Curie Point of the principal phase in most samples appears to be near 600° C. Forty-seven per cent of the normal sites, 55 per cent of the reversed sites and 70 per cent of the intermediate sites show a low critical blocking temperature component.

3.4. *Analysis of mean directions of magnetization*

Ideally palaeomagnetic studies yield two anti-parallel sets of normal and reversed directions for which separate means can be determined. In the present study the situation is complicated because of the large number of ' intermediate ' directions (Fig. 2(a)). There is no bipolar analogue of the Fisher (1953) distribution which is unimodal, and we have to recognize the dual polarity in these data and transpose the ' reversed ' vectors for Fisher analysis. The first step is to contour the raw data on an equal-area counting net described by Kalsbeek (1963) (Fig. 5). The two principal maxima are near $D = 48°$, $I = -49°$ on the upper hemisphere and $D = 216°$ $I = +50°$ on the lower hemisphere. The mean axis of magnetization may therefore be taken as $D = 222°/42°$ $I = \pm 49°$ which is the best estimate derivable from the data and corresponds to palaeomagnetic poles: 4° S 141° E and 4° N 321° E. However for comparison with other palaeomagnetic data it is desirable to calculate Fisher parameters. The data must first be ' normalized ' to a single polarity, and this is done by projecting on to the hemisphere centred on $D = 42°, I = -50°$. Fisher analysis now gives a refined mean $D = 34°$, $I = -45°$, $k = 3{\cdot}1$ but this is not strictly valid for two reasons: firstly, the cut-off at $\theta = \pi/2$ from the mean is artificial and violates the Fisher distribution, and secondly the distribution within the hemisphere is non-Fisherian, having many more observations in the range $70° < \theta < 90°$ than is attributable to chance (Watson & Irving 1957). The anomalous observations (here called ' intermediate ', Section 3.5) have been removed by stages until a χ^2-test indicates that a Fisherian distribution is satisfied. This stage is reached when $N = 34$ and $k = 12{\cdot}0$

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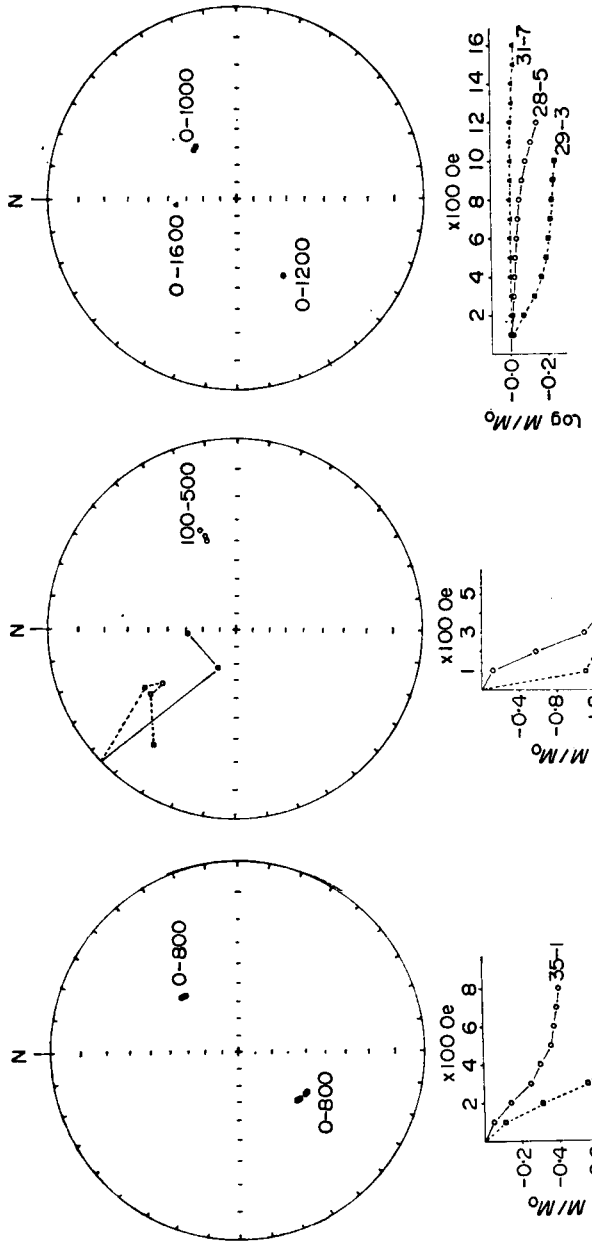


FIG. 3. Directions of magnetization and alternating field demagnetization curves for (a) two specimens showing a near-exponential fall off in their magnetic moments, (b) two specimens showing a sharp drop in their magnetic moments, and (c) three specimens of high coercity showing only slight fall in their moments with applied demagnetizing field.

Table 1

Site statistics after tilt-correction

Site	Grid ref.*	Total NRM: No. of samples	R	D,	I	Peak AF (Oe)	AF cleaned: No. of samples	R	k	α_{95}	D,	I	VGP °E,	VGP °N	Group†
3.	113085	5	4·99	75,	−35	(300)‖	4	3·77	13·2	26·3	(68,	−52)‖	111,	−8	N
4.	105118	7	—	—	—	400	4	3·74	11·7	28·1	220,	−57	285,	−57	
6.	092424	6	5·91	222,	+57	300	5	4·93	53·5	10·6	223,	+45	319,	0	R
8.	208316	7	6·16	162,	+46	300–400	6	5·82	28·1	12·9	157,	+39	18,	−9	
9.	223317	7	5·50	354,	+36	300	5	4·70	13·4	21·7	52,	−36	129,	+2	N
10.	258317	5	4·67	4,	+40	300	4	3·89	27·4	17·9	334,	−38	202,	+9	
11.	115212	5	3·65	42,	−22	300	3	3·00	—	—	39,	−49	143,	−2	N
12A.	115210	5	4·71	175,	+1	400	2	2·00	—	—	175,	−31	4,	−50	B
12B.‡	115210	5	4·90	172,	−4	300–400§	3	2·96	—	—	174,	−15	5,	−41	B
13.	195281	4	3·97	19,	+47	300	3	2·99	—	—	45,	+32	117,	+38	A
14.‡	195281	3	2·96	44,	+24	300§	2	1·99	—	—	55,	0	116,	+18	A
15.	243288	5	4·86	72,	+45	300	3	2·96	—	—	81,	+40	81,	+24	A
16.	241281	5	—	—	—	300	3	2·97	—	—	137,	−63	56,	−71	B
17.	420370	6	5·47	35,	−22	(300)‖	5	4·86	28·6	14·6	(355,	−60)‖	142,	+16	N
18.	492351	6	4·51	69,	+16	400	4	3·60	7·5	36·0	139,	−32	52,	−40	B
19.	486351	6	5·73	162,	−6	300	5	4·69	12·8	22·2	144,	−35	48,	−44	B
20.	628496	5	4·20	300,	+61	400	4	—	—	—	—	—	—	—	
21.	632493	6	5·79	307,	+75	300	4	3·95	55·9	12·4	181,	+71	357,	+22	
22.	755647	5	4·96	300,	+61	300	4	3·96	76·5	10·6	277,	+56	285,	+34	
23.	752647	3	—	—	—	500	2	1·99	—	—	25,	−20	151,	+20	N
24.	797669	6	5·80	41,	−58	600	5	4·90	41·6	12·0	72,	−54	122,	−20	N
26.	880797	5	4·40	306,	+64	300	4	—	—	—	—	—	—	—	
27.	879797	5	4·89	244,	+73	300	4	3·82	16·6	23·2	210,	+66	338,	+18	R
28.	878797	5	4·95	235,	+35	300	4	3·69	9·7	31·1	232,	+40	312,	+1	R
29.	877797	6	5·95	46,	−53	900	5	4·95	84·6	8·4	43,	−54	143,	+8	N
30.¶	877797	7	3·49	—	—	400	5	2·03	—	—	—	—	—	—	
31.	877872	10	9·81	332,	−62	300	6	5·90	49·7	9·6	340,	−59	193,	−8	
32.‡	887872	4	3·85	315,	−52	300§	4	3·87	23·0	19·6	314,	−50	216,	−6	
33.	882832	4	3·94	24,	−27	300	2	2·00	—	—	52,	−51	136,	−8	N
34.	879830	5	4·98	27,	−26	600	3	3·00	—	—	30,	−32	148,	+12	N
35.	842733	3	2·68	16,	−15	0	3	2·68	—	—	16,	−15	160,	+24	N
36.	842733	3	2·97	219,	+54	0	3	2·97	—	—	219,	+54	326,	+7	R
37.	799670	5	4·87	173,	−39	200	3	2·94	—	—	174,	−39	8,	−55	B
38.	783663	5	—	—	—	300	4	3·99	484·4	4·2	53,	−75	150,	−35	N

39.	785663	5	4·99	168,	−24	500	3	3·00	—	—	167,	−23	15,	−44	B
40.	778747	4	3·99	232,	+30	200	3	3·00	—	—	234,	+30	307,	−5	R
41.	760801	6	5·87	208,	+42	100	5	4·95	63·6	9·7	218,	+44	323,	−1	R
42.	760800	5	4·73	186,	+46	100	3	2·98	—	—	213,	+51	330,	+2	R
43.	767809	5	4·94	51,	−50	300	3	2·97	—	—	52,	−50	134,	−8	N
44.	247354	5	4·73	29,	−17	700	4	3·61	7·7	35·5	30,	−14	144,	+22	N
45.	220258	5	4·98	64,	+51	200	2	2·00	—	—	69,	+51	84,	+37	A
46.	212259	4	4·00	44,	−65	(100)‖	3	2·86	—	—	(71,	−70)‖	148,	−20	N
47.	207260	5	4·86	103,	+62	700	4	3·90	29·0	17·4	105,	+42	61,	+12	A
48.	203260	5	4·71	296,	+79	500	5	4·50	8·1	28·7	176,	+44	0,	−8	R
49.	202263	6	5·73	54,	+11	500	5	4·96	101·2	7·6	56,	−12	119,	−13	N
50.	207297	6	5·76	4,	+30	500	4	3·52	6·2	40·2	18,	0	115,	+14	N
51.	202291	5	4·34	15,	+34	500	3	2·99	—	—	56,	+10	113,	+22	A
52.	204292	4	3·94	191,	+42	700	3	2·99	—	—	186,	+27	351,	−19	R
53.	203278	5	4·95	51,	+47	700	4	3·96	74·9	10·7	81,	+6	93,	+7	A
54.	284545	5	4·93	22,	−38	700	3	2·98	—	—	26,	−45	154,	+4	N
55.	256537	6	—	—	—	300	4	3·53	6·4	39·4	32,	−7	142,	+24	N
56.	203278	4	3·83	75,	+28	700	2	1·99	—	—	85,	+14	87,	+9	A
57.	207282	5	4·46	43,	+78	700	4	3·94	49·2	13·2	72,	+5	101,	+12	A
58.	506502	6	4·24	301,	+54	500	4	3·52	6·3	39·9	259,	+42	299,	+14	R
59.	508496	6	5·76	237,	+19	700	4	3·96	86·3	10·0	236,	+8	299,	−14	R
60.	537447	5	4·82	246,	+23	700	3	2·98	—	—	234,	+5	300,	−17	R
61.	548452	5	4·86	156,	+56	300	3	—	—	—	—	—	—	—	
62.	740708	8	5·43	100,	+61	500	4	3·68	9·5	31·4	116,	+53	48,	+15	A
63.	732568	5	4·95	214,	+64	700	4	3·99	418·6	4·5	212,	+51	331,	+2	R
64.	696525	7	—	—	—	—	—	—	—	—	—	—	—	—	
176.	065419	6	—	—	—	800	5	4·59	9·7	25·8	93,	+48	68,	+22	A
177.	063413	5	—	—	—	800	4	3·71	10·5	29·8	332,	+14	212,	+36	
105.	074425	6	—	—	—	300	5	4·88	32·2	13·7	263,	−53	252,	−31	
174.	276541	5	—	—	—	400	5	4·70	13·2	21·8	45,	−28	133,	+9	N
107.	277543	9	8·70	32,	−42	300	9	8·83	47·6	7·5	31,	−45	150	+3	N
173.	280542	6	3·98	94,	−56	500	4	—	—	—	—	—	—	—	
133.	767810	9	8·93	41,	−51	300	8	7·93	94·9	5·7	38,	−48	144,	−2	N

Notes:

* Grid references are in square NO of the UK National Grid.

† The sites are classed as normal (*N*), reversed (*R*), or as one of two intermediate groups (*A* and *B*) as explained in the text. For unclassified sites this column is left blank.

‡ Sediments.

§ Thermally cleaned. This column gives temperature in °C.

‖ Brackets denote that total NRM was used in the final analysis.

¶ Conglomerate.

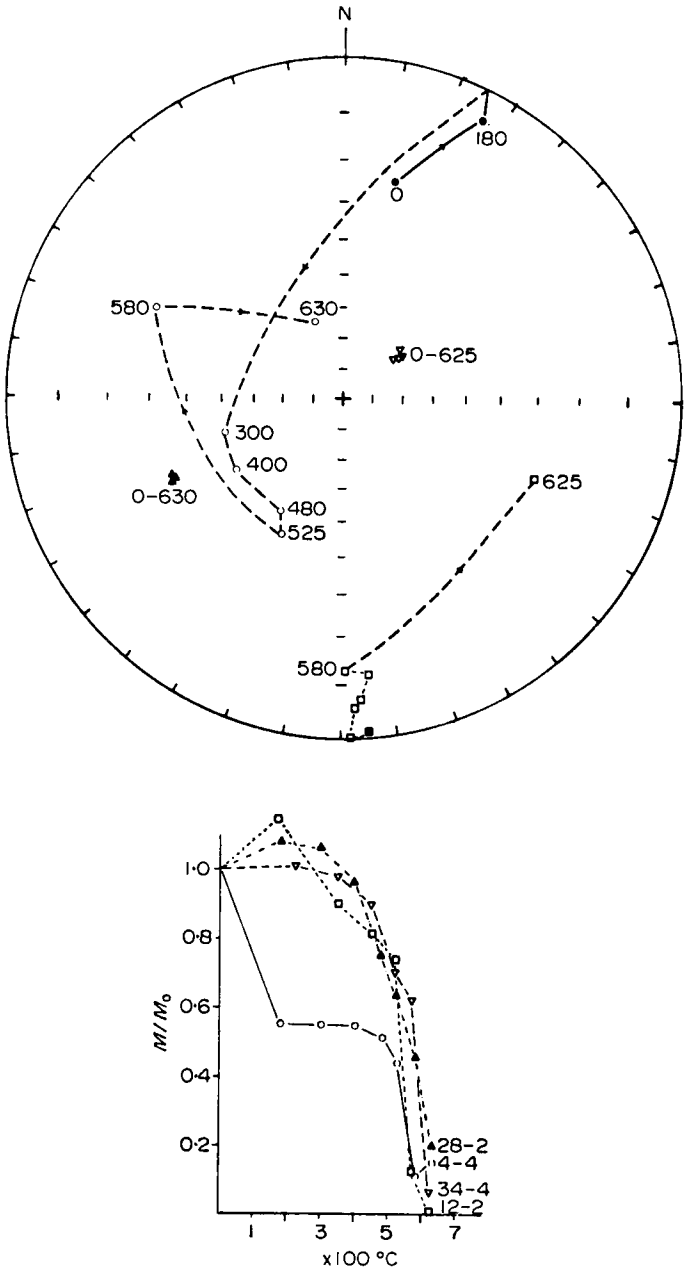


FIG. 4. Directions of magnetization and demagnetization curves for four thermally demagnetized samples.

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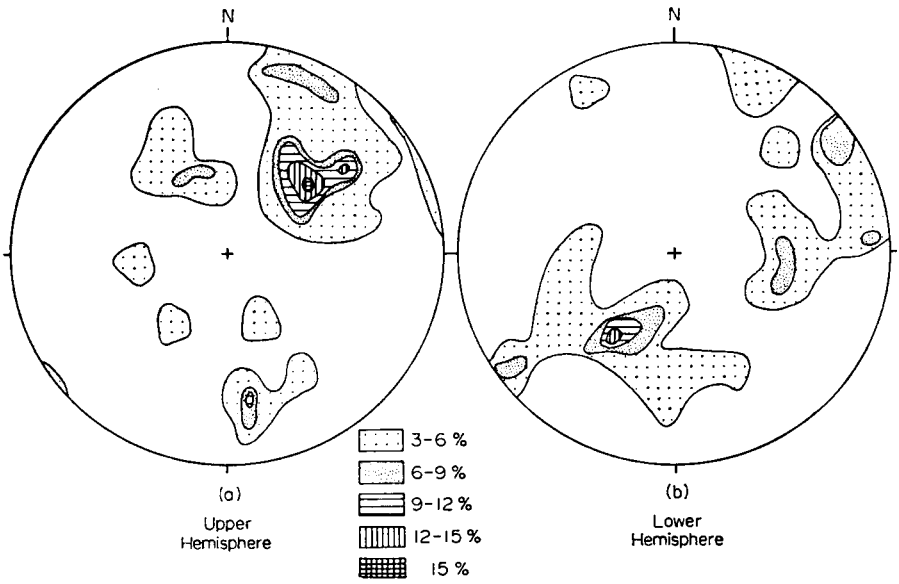


FIG. 5. Contoured Kalsbeek plot of directions of magnetization on (a) upper, and (b) lower hemispheres after magnetic or thermal cleaning and tilt correction.

and these directions are regarded as of ' normal ' and ' reversed ' polarity and it is their analysis which provides the best Fisherian estimate of the typical magnetization in these rocks (Table 3).

3.5. *Polarity groups*

The ambiguity in defining polarity when the Virtual Geomagnetic Poles (VGPs) are grouped close to the present equator may be resolved by reference to the polar wander curve for the region (McElhinny *et al.* 1968). On this basis the 21 north-easterly upward sites are ' normal ' and the 13 south-westerly downward sites are ' reversed ' Thus only 34 of the 61 significant site mean directions are classified as normal or reversed by the foregoing procedure. The remaining 27 site mean directions are random ($R = 7.72$) with more than 5 per cent probability when treated as a group. However, they actually comprise two groups (Fig. 5)—Group A including 11 sites with easterly declination and shallow positive inclination, and Group B of seven sites with southerly declination and moderate negative inclination—superimposed on a thoroughly random background of nine unclassified sites ($R = 3{\cdot}09$).

4. Geological evidence for the age and origin of remanence

There are four lines of geological evidence that the remanence in all these rocks, with the possible exception of some of the ' unclassified ' sites, is of Lower Devonian age. Their stratigraphic distribution of all polarity groups (*N*, *R*, *A*, *B* and probably the unclassified one) is consistent with a polarity sequence (Section 4.4). Fold tests (Section 4.1.) indicate that the remanence at *N*, *R*, *A* and *B* sites pre-dates Middle Devonian folding. A conglomerate test (Section 4.2.) demonstrates primary remanence at some *N* and *R* sites. Igneous contact tests (Section 4.3.) confirm primary remanence in sites of Groups *A* and *B*, and at two sites in the ' unclassified ' group of Section 3.5, although these two particular sites may not be typical of that group.

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4.1. *Fold tests*

Comparison of Fisher precision estimates, k, before and after allowing for local folding (Table 2) shows that both at the NRM level and especially after magnetic cleaning, the bulk of the remanence in *N*, *R*, *A* and *B* sites pre-dates the folding which is well dated within the Middle Old Red Sandstone interval (MacGregor & MacGregor 1948; Armstrong & Paterson 1970). In most cases the improvement in precision is significant with better than 95 per cent confidence (Cox 1969). McMurry (1968, 1970) obtained a similar result from the *N* and *R* groups, but this new analysis provides striking confirmation. The apparent indication in this test that the *A* and *B* intermediate groups are also older than the folding and hence that their designation as 'intermediate polarity' is justifiable, should be treated with caution—the improvement in precision after 'correcting' for tilt may be a consequence of the analysis, since they were themselves defined in terms of tilt-corrected directional groups.

4.2. *Conglomerate test*

At Crawton Bay seven samples were collected (site 30) from lava pebbles in a conglomerate immediately overlying a section of three lava flows (Fig. 6). The lowest lava (site 26) is magnetically unstable to AF demagnetization and perhaps Embleton's samples from this section all came from this flow, because the properties he observed are not consistent with those in the higher flows. Stubbs (1958) found a reversed total NRM in this flow included in Fig. 6. Sites 27 and 28 from the two units of the middle flow have reversed polarity. The top flow (site 29) is normal. These flows show a range of AF stability between the types illustrated in Fig. 3(a) and (c) reflected in their varied oxidation states. The lavas and the conglomerate pebbles are of distinctive macroporphyritic basalt, and since this is the only locality in the whole Lower Old Red Sandstone sequence where such lavas are known the provenance of the pebbles can only be local. Therefore our finding that the conglomerate pebbles

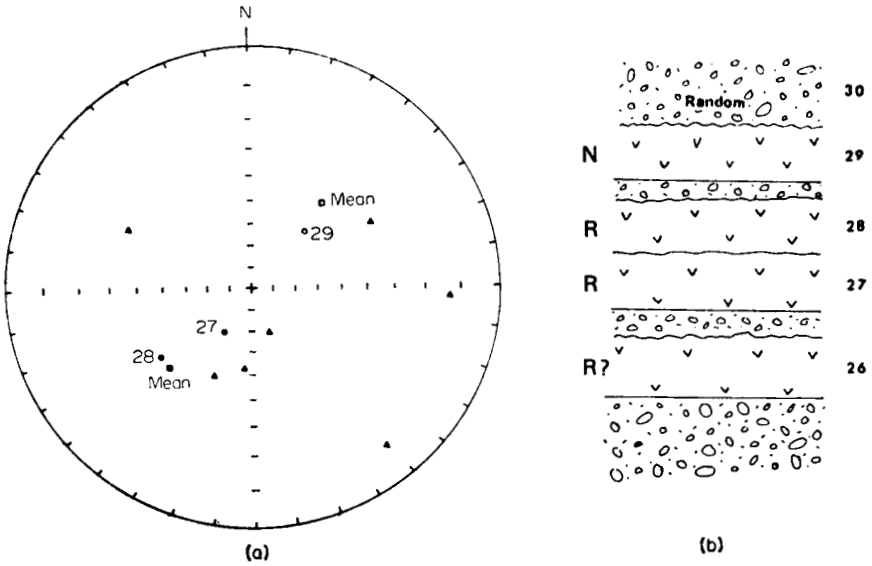


FIG. 6. (a) A plot of site mean directions from the Crawton Bay volcanics (circles) together with cleaned directions of magnetization for seven samples (site 30) from pebbles in an overlying conglomerate (triangles). The squares show the accepted Lower Devonian magnetic field direction. (b) Diagrammatic representation of the stratigraphic distribution of sites at Crawton Bay.

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Table 2

Fold tests

	Total NRM					AF cleaned				
	Number of sites	Precision, k *in situ*	Precision, k tilt corrected	Significance of improvement	D, I tilt corrected	Number of sites	Precision, k *in situ*	Precision, k tilt corrected	Significance of improvement	D, I tilt corrected
Normal group	17	3·6	6·2	>90%	33, −28	21	5·6	11·9	99%	39, −37
Reversed group	13	6·5	6·0	None	228, +47	13	4·9	11·5	>95%	221, +41
N and *R* combined	30	4·2	5·7	Low	38, −37	34	5·5	12·0	>99%	40, −39
Group *A*	10	8·8	9·7	Low	55, +50	11	7·1	9·0	Low	76, +29
Group *B*	6	3·2	3·8	Low	158, −11	7	12·5	17·5	Low	162, −35

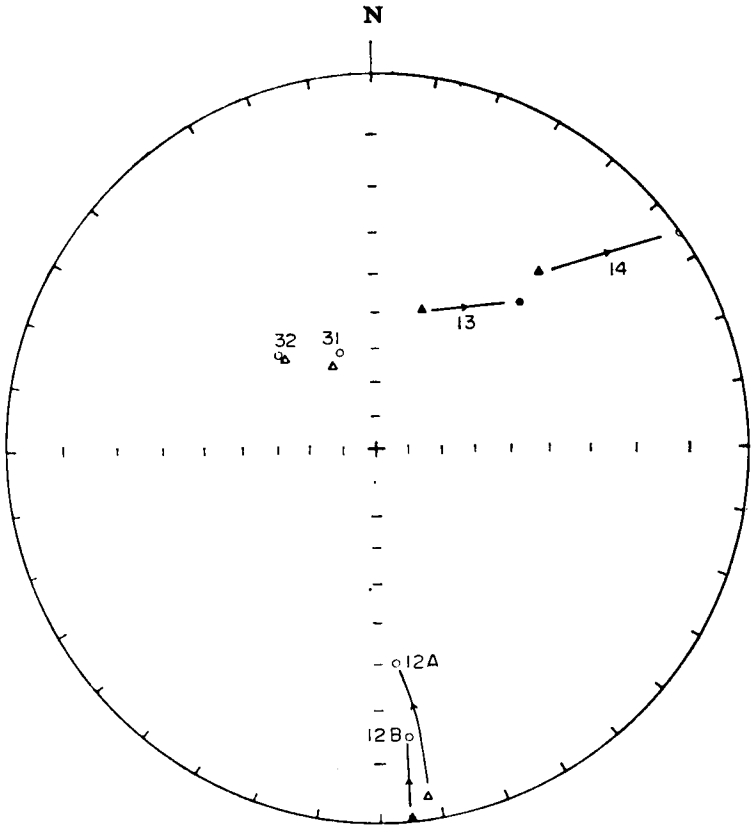


FIG. 7. Directions of magnetization at the three pairs of contact test sites before (triangles) and after AF or thermal cleaning (circles).

had similar magnetic properties and stability to the lavas (sites 27–29) and differed *only* by being randomly magnetized, is convincing evidence that the remanence is older than the conglomerate and hence dates from the cooling of the lavas themselves. Moreover the conglomerate *pebbles* show varying degrees of oxidation, so that this oxidation is itself virtually primary and may determine the NRM intensity.

4.3. *Igneous contact tests*

Three pairs of sites have been sampled in lava and immediately underlying sediments and in all three cases the direction of stable remanence agrees (Fig. 7) suggesting the absence of self-reversal of polarity in all of these rocks. The pair of sites 13 and 14 belong to polarity group *A*; and 12A and 12B to polarity group *B*. The contact tests provide strong indication that these intermediate groups are genuine records of intermediate polarity. Sites 31 and 32 have been ' unclassified ' in Section 3.5. Their discrepancy from the normal group is mainly one of declination, which can be accommodated if they have been rotated 45–60° anticlockwise in the plane of the Highland Boundary Fault (assumed to be vertical and strike N55° E). The geological evidence suggests that such a rotation is excessive and even after it is effected there remains a discrepancy in direction of 25–40°. These considerations and the contact test suggest the possibility that sites 31 and 32 may represent a true Downtonian field direction. This is a special case; the other ' unclassified ' directions may not necessarily be explained in this way and it cannot be assumed that they are primary.

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FIG. 8. The stratigraphic distribution of palaeomagnetic sites in the Lower Old Red Sandstone rocks of the Strathmore region. The stratigraphic columns are based on Armstrong & Paterson (1970) and their general location is given by the bracketed letters in Fig. 1. The column on the right-hand side of this figure gives a probable history of the magnetic field direction for the interval covered by the collection. The indication ?*R* refers to Stubbs (1958) and ?*B* to our intermediate observations whose stratigraphic levels are uncertain.

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4.4. *Magnetic stratigraphy*

The stratigraphic position of each of our palaeomagnetic sites is shown (Fig. 8) with the results of McMurry (1968) and Embleton (1968) incorporated. The stratigraphic columns are taken from Armstrong & Paterson (1970) and their approximate locations are given in Fig. 1. Their correlation, together with the composite magnetic polarity sequence in Fig. 8 is due to I.B. Paterson. One of the principal contentions of Armstrong & Paterson (1970) is that the Collace sequence is approximately the same age as the St Cyrus lavas. On this basis it is apparent that the majority of our *A* and *B* polarity groups are confined to a short stratigraphic interval near the middle of the Arbuthnott Group; but they are separated by intervals during which *N* and *R* directions pertained. This high incidence of *A* and *B* directions over this brief interval may conversely support the correlation of the Collace and St Cyrus sections which is based purely on stratigraphic evidence.

Group *A* comprises nine sites in lavas, one in a baked sedimentary contact and one site in the Highland Border Ignimbrite. All of these except the last (176) occur on one traverse in the vicinity of Collace in the Ochil Volcanic Formation. All seven sites in group *B* belong to the Arbuthnott Group and occur over a distance of more than 65 km across the Sidlaw anticline; these results show that there are at least two, and perhaps four, *B*-polarity intervals in this sequence.

Above this anomalous sequence there is evidence of normal polarity at the very top of the Arbuthnott Group; but after that there are only isolated observations of *R* and later *A* polarity within the Garvock Group. In the Lower Arbuthnott Group there are 15 sites in 13 separate localities. These observations are compatible with a simple *N–R–N* sequence. The earlier of these two *N*-epochs probably began in late Crawton Group times and was preceded by an *R*-epoch recorded in the section of Crawton Bay. Prior to that there are only three observations, the youngest of which has *N* polarity and the oldest (Stonehaven) sites have anomalous directions discussed in Section 4.3.

The principal conclusion is that the overall sequence of events is a relatively long period *R–N–R–N* alternation, but that a series of geomagnetic events occurred at the middle of the Arbuthnott Group when the magnetic pole moved rapidly to occupy and reoccupy several divergent positions over a short interval. The most probable sequence of observed polarities is *N–B–A–R – B–N–(B?)–A–R*. As the time coverage may not be complete, this suggests a possible underlying rhythmic pattern *N–B–A–R–A–B–N–B–A–R*. In this scheme, the first deviation from definite polarity (*N* or *R*) takes the form of a declination change; the next change is principally a switch of inclination; finally declination swings into line to complete the reversal. For some unknown reason the magnetic field appears to have lingered in the intermediate positions *A* and *B* during polarity transitions.

Since the total NRM within-site mean directions are generally consistent for these lavas and the demagnetization results (Section 3.2) show that most samples have moderate to high stability, it is pertinent to re-examine the total NRM results of Stubbs (1958). His results from Crawton Bay suggest that his lowest and third flows have reverse magnetization and the top flow is normal, a polarity transition which agrees with our work if Stubbs' ' third ' flow is in fact the upper unit of the middle flow. Two of three samples from St Cyrus (Stubbs' site 7) have reversed directions and are near our ' unclassified ' site 22, suggesting that a reversal of magnetization takes place within the St Cyrus lavas between these sites and our normal site 23. Stubbs' site 8 at Arbuthnott House has a direction similar to our site 43 and his site 9 is reversed confirming this polarity transition between the top of the Arbuthnott and the lower part of the Garvock Groups. A further reversal is suggested from Stubbs' sites at Knock Hill: his sites 12, 13 and part of 14 with normal directions, are underlain by a reversed horizon of the remainder of site 14 and site 15. In the section south of Montrose Stubbs' sites 16 and 18 have reversed directions in common with sites 63

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and F1, but a transition to normal polarity above this is demonstrated by McMurry's sites U1 and U2.

The Highland Border ignimbrites occur as small and scattered outcrops which have been correlated on petrological similarity by several workers notably Allan (1928) and Paterson & Harris (1969). At the two most westerly outcrops, at Dunkeld and Lintrathen, the ignimbrites rest unconformably on Dalradian metasediments but at Glenbervie, near Stonehaven, the ignimbrite occurs within the Lower Old Red Sandstone sequence at an horizon which is correlated with the Crawton Group by Armstrong & Paterson (1970). The two lower sites (174 and 107) at Lintrathen and the single site at Glenbervie are normally magnetized, consistent with correlation of these horizons and their identification with the top of the Crawton section. The remaining four sites have stable remanence (or, in the case of site 173, total remanence) in four different quadrants. Three of these sites are in the Dunkeld district where Paterson & Harris (1969) recognize multiple ignimbrite formation. They all have intensities of magnetization considerably lower (in the range $0{\cdot}5\times10^{-7}$ to $1{\cdot}0\times10^{-5}$ Gauss) than the three normally magnetized ignimbrites ($1{\cdot}0\times10^{-5}$ to $5{\cdot}0\times10^{-5}$ Gauss) although susceptibilities are comparable, suggesting that they may have acquired their magnetism during a period of polarity transition. Taken together these sites would be regarded as randomly magnetized and it may not be correct to single out just one of them although site 126 at Dunkeld does lie close to our '*A*' group. The palaeomagnetic results therefore do not provide any direct evidence for correlating this horizon with the top of the Crawton Group. McMurry's (1968) anomalous sites C1 and C2 could be classified as belonging to Group *A* and these sites are stratigraphically below the section at Crawton (sites 26–30, McMurry's C3, C4) and indicate that there may be a further *A* Group horizon below the top of the Arbuthnott Group.

Smith (1967) recognized the occurrence of the directions which we call 'intermediate' in the Ochil Volcanic Formation near Stirling, but he was inclined to ascribe them to a secondary origin. But neither group is magnetized in a direction expected for any post-Devonian time. Group *B* is closest to late Palaeozoic directions, and Group *A* closest to Jurassic directions but in neither case is the match precise; nor is there any geological evidence for remagnetization processes of either of those ages, and certainly no hypothesis has been suggested for selective remagnetization over such a wide area. On the contrary, our recognition of their stratigraphic distribution, plus the results of igneous contact tests, refutes this view and confirms these directions as of Lower Devonian age. Creer & Embleton (1967) and Embleton (1968) discarded many of their observations without explanation and it is permissible to wonder whether some might have fallen into this category. McMurry (1968) found one site (N1) which we would term 'unclassified'. Its direction is not drastically different from the normal polarity group and McMurry included it in his analysis but, for the sake of consistency it has been excluded here, Table 3.

5. Rock magnetic properties and the origin of remanence

The lavas have intensities of magnetization in the range of 10^{-2} to 10^{-5} Gauss (Fig. 9(a)) with a logarithmic mean intensity of $3{\cdot}5\times10^{-4}$ Gauss. The logarithmic mean intensity of the 'intermediate' sites is $5{\cdot}0\times10^{-4}$ Gauss. The susceptibilities have been measured with an AC bridge (Collinson & Molyneux 1967). The normal and reversed sites have a logarithmic mean susceptibility of $4{\cdot}0\times10^{-4}$ Gauss/Oersted (Fig. 9(b)) and the 'intermediate' sites have a log.-mean susceptibility of $3{\cdot}5\times10^{-4}$ Gauss/Oersted. The AF coercivity spectra of NRM are compared in Fig. 10 with that of a TRM imposed by cooling from 700° C in a field of 0·42 Oe in the laboratory. Heating often affects the specimens drastically; two exploded in the furnace but in less extreme cases chemical alteration takes place and this is invariably accompanied by a change in susceptibility. This accounts for all the specimens which show erratic

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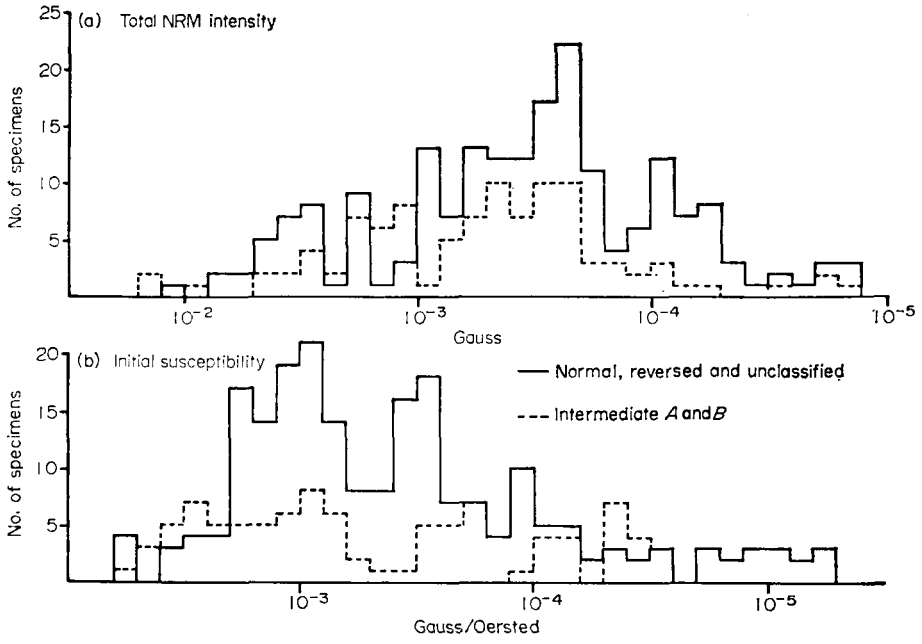


FIG. 9. Frequency distribution of (a) intensities of magnetism and (b) susceptibilities of the Strathmore lavas grouped into sites falling into the intermediate '*A*' and '*B*' and the normal, reversed and unclassified groups.

variation of NRM/TRM ratio as a function of applied alternating field and also for site 29. Five examples of constant NRM/TRM ratio are in the range 0·1–0·3 and they are from two reversed, one normal, and two ' unclassified ' sites. Because of the limitations of this elementary experiment, it is merely concluded that there is no evidence that the character of the magnetization is different between ' normal ', ' reversed ', and ' unclassified ' sites or that their remanences were acquired under different geomagnetic field intensities. The evidence points to an ancient field intensity an order of magnitude less than its present value.

These properties do not correspond with the common variation of remanence in mixed polarity sequences in two respects. First, the incidence of intermediate polarity (our *A*, *B* and ' unclassified ' groups) is much greater than is usually observed, even in eras of frequent polarity change. This could conceivably be accounted for during the time interval covered by the present study if the frequency of field reversal was greater than has been found during more recent geological time. Alternatively the contention that intermediate directions are records of geomagnetic polarity transition may be erroneous. There is no evidence (except possibly in the Highland Border Ignimbrite) that the intermediate polarities correlate with low remanent intensity and hence low ambient geomagnetic field, as is commonly the case for rocks formed during geomagnetic polarity transitions.

6. The palaeomagnetic pole position

The normal and reversed mean directions of magnetization combine to provide the best estimate of the typical local Lower Devonian geomagnetic field (Table 3). This result is almost identical with that of McMurry (1968, 1970) which is listed in Table 3 together with the results of Stubbs (1958) and Embleton (1968). The present study forms the culmination of intensive palaeomagnetic work on these rocks over two decades, and the moment is opportune to consolidate the results of the various

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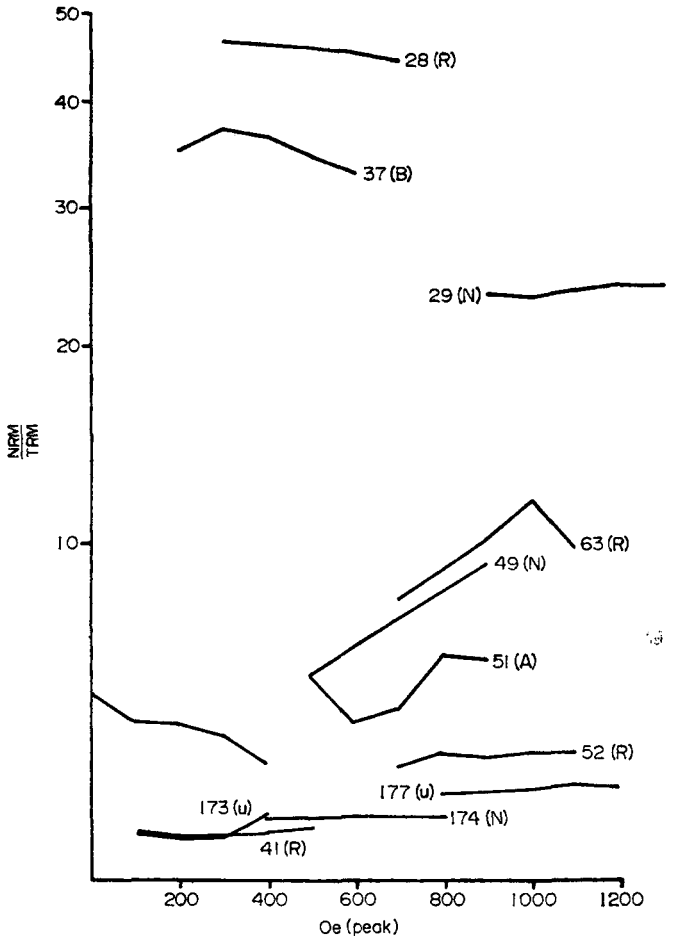


FIG. 10. NRM/TRM ratios as a function of applied alternating field for ten lava and two ignimbrite sites.

workers. Composite results are given in Table 3, taking account of duplication of sampling sites. The result of Stubbs is based on total NRM measurements only. Eight of his sites are covered by demagnetization work of McMurry, and acceptance of the remaining eight would make little difference to the composite result; for consistency, the best estimate depends entirely on demagnetized results. Embleton's results, like Stubbs', give a more easterly declination than is typical. This discrepancy is probably a consequence of the small number of sampling sites, although there are some sites in our study with large declinations (3, 24, 58). Because only four of Embleton's sites are included in the composite calculation they alter the mean by only a few degrees. On this basis the palaeomagnetic pole is best estimated as 4° N, 140° E ($d\psi$, $d\chi$ = 4°, 7°) from 50 sites.

This composite result is deliberately confined to the Midland Valley. Data of Embleton (1968) from the Lorne Plateau and of McMurry (1968, 1970) from Glencoe will be discussed separately in Paper VI although they were incorporated with Midland Valley data by the original authors.

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Table 3

Summary of normal and reversed palaeomagnetic directions in Lower Devonian lavas from the Midland Valley of Scotland.*

	Partial demagneti-zation†	Number of sites‡ Normal	Reversed	Total	k	α_{95}	D, I	Palaeomagnetic North Pole		$d\psi$, $d\chi$
Stubbs (1958)	none	8 (5)	8 (3)	16 (8)	25	8	46,– 54	10° S	141° E	8, 11
Embleton (1968)	*a* (*t*)	3	2 (1)	5 (1)	14	21	48,– 37	3 N	133 E	15, 25
McMurry (1968)	*a*	6 (1§)	9 (2)	15 (3)	16	10	37,– 44	2 N	144 E	8, 12
present study	*a* (*t*)	21	13	34*	12	7	40,– 39	5 N	140 E	5, 9
All results		32	26	58	14	5	42,– 42	2 N	139 E	4, 6
All results based on partial demagnetization		29	21	50	13	6	41,– 40	4 N	140 E	4, 7

* Includes three sites in sediments adjacent to lavas.

† *a* denotes alternating field, *t* denotes thermal demagnetization.

‡ Numbers in brackets are number of sites superseded by later studies, taking them in the order in which entries are listed in this Table.

§ One site regarded as ' unclassified ' by the present authors, see text.

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We would like to thank Mr I. B. Paterson of the Institute of Geological Sciences, Edinburgh, for his very considerable help in locating suitable sampling sites and interpreting geological aspects of the results. Several sections of this paper, and in particular the interpretation of the magnetic stratigraphy, have also benefited greatly from his comprehensive criticism. Dr J. C. Briden kindly provided much useful advice in interpreting results and improving the manuscript. This work was carried out while one of us (J.T.S.) held a scholarship from the Iraqi Ministry of Oil, who financed the field work. We are further indebted to Professor R. L. Wilson of Liverpool University for the loan of field drilling equipment.

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